Project: Molecular dynamics simulation of different condensed phases

Objective: To study the structure and dynamic properties of a Lennard-Jones (LJ) system in different condensed phases (a solid, a liquid and a gas).

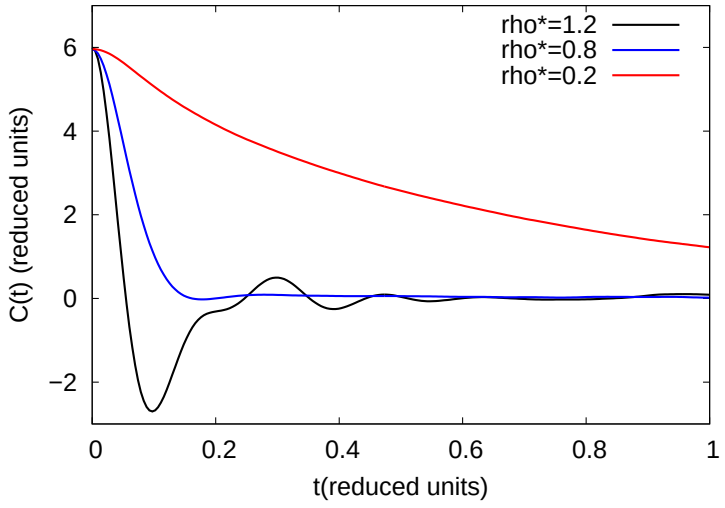
Outline

1. Write a MD code for a (N,V,T) simulation of a Lennard-Jones (LJ) system. Use N=500 atoms. Remember that the LJ potential is of the form: $U_{LJ}(r) = 4\varepsilon[(\sigma/r)^{12} - (\sigma/r)^{6}]$.
2. Create an initial set-up of positions from the sites of a fcc (*face centred cubic*) lattice. (HINT: consider a fcc lattice as the superposition of 4 single cubic lattices.)
3. Create an initial set-up of velocitie from a Maxwell distribution. Remember that the velocity of the center of mass of the system must be zero.
4. Run a MD simulation for argon at $T^* = 2.0$ and $\rho^* = 1.2$. Re-scale the temperature if necessary. Compute the interaction energy U^*, the pressure P^*, the radial distribution function $g(r)$,the velocity autocorrelation function $C(t)$ and the mean square displacement $r^2(t)$ of the system.
5. Repeat the simulation for the following state points: $T^* = 2.0$; $\rho^* = 0.8$, and $T^* = 2.0$; $\rho^* = 0.2$, and compare with the previous results.

Lennard-Jones parameters

atom	mass(uma)	ϵ/k_B (K)	σ ($\AA$)
Ar	39.948	119.8	3.405

Some results to compare with:



Velocity autocorrelation functions for different densities.

Questions

1. **Show the equivalence between the leap-frog Verlet and the velocity Verlet algorithms.**

2. Define the reduced units used in your simulations. Translate from reduced units to real units the values of T^*, ρ^*, U^* and P^*.

3. Given an interpretation for the observed initial value of the velocity autocorrelation functions $C(0)$.

4. Discuss separately the short and long time behaviour of the $r^2(t)$ functions. Compute the self-diffusion coefficients D in reduced and in real units.